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# Student Result Analysis System
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```
import tkinter as tk
from tkinter import messagebox
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
# Data storage
```

```
data = []
```

```
# Function to add student data
```

```
def add_data():
    name = name_entry.get()

    cpp = int(cpp_entry.get())
    python = int(python_entry.get())
    java = int(java_entry.get())
    c = int(c_entry.get())
    os = int(os_entry.get())

    data.append([name, cpp, python, java, c, os])

    messagebox.showinfo("Success", "Student data added")

    name_entry.delete(0, tk.END)
    cpp_entry.delete(0, tk.END)
    python_entry.delete(0, tk.END)
    java_entry.delete(0, tk.END)
    c_entry.delete(0, tk.END)
    os_entry.delete(0, tk.END)
```

```
# Function to analyze results
```

```
def analyze():
    if len(data) == 0:
        messagebox.showwarning("No Data", "Enter student data first")
        return

    df = pd.DataFrame(data, columns=["Name", "C++", "Python", "Java", "C", "OS"])

    marks = df[["C++", "Python", "Java", "C", "OS"]].values

    avg = np.mean(marks)
    highest = np.max(marks)
    lowest = np.min(marks)
```

```
lowest = np.min(marks)

result_label.config(
    text=f"Average Marks: {avg:.2f}\nHighest Marks: {highest}\nLowest Marks: {lowest}"
)

# Bar chart
df.set_index("Name").plot(kind="bar")
plt.title("Student Marks")
plt.ylabel("Marks")
plt.show()

# GUI Window
root = tk.Tk()
root.title("Student Result Analysis System")
root.geometry("400x450")

# Labels and Entries

tk.Label(root, text="Student Name").pack()
name_entry = tk.Entry(root)
name_entry.pack()

tk.Label(root, text="C++ Marks").pack()
cpp_entry = tk.Entry(root)
cpp_entry.pack()

tk.Label(root, text="Python Marks").pack()
python_entry = tk.Entry(root)
python_entry.pack()

tk.Label(root, text="Java Marks").pack()
java_entry = tk.Entry(root)
java_entry.pack()

tk.Label(root, text="C Marks").pack()
c_entry = tk.Entry(root)
c_entry.pack()

tk.Label(root, text="OS Marks").pack()
os_entry = tk.Entry(root)
os_entry.pack()

# Buttons
tk.Button(root, text="Add Student", command=add_data).pack(pady=10)
tk.Button(root, text="Analyze Result", command=analyze).pack(pady=10)

# Result Label
result_label = tk.Label(root, text="")
result_label.pack(pady=20)

root.mainloop()
```

Student Result Analysis System

Student Name

Subrahmanya KR

C++ Marks

85

Python Marks

96

Java Marks

75

C Marks

63

OS Marks

72

Add Student

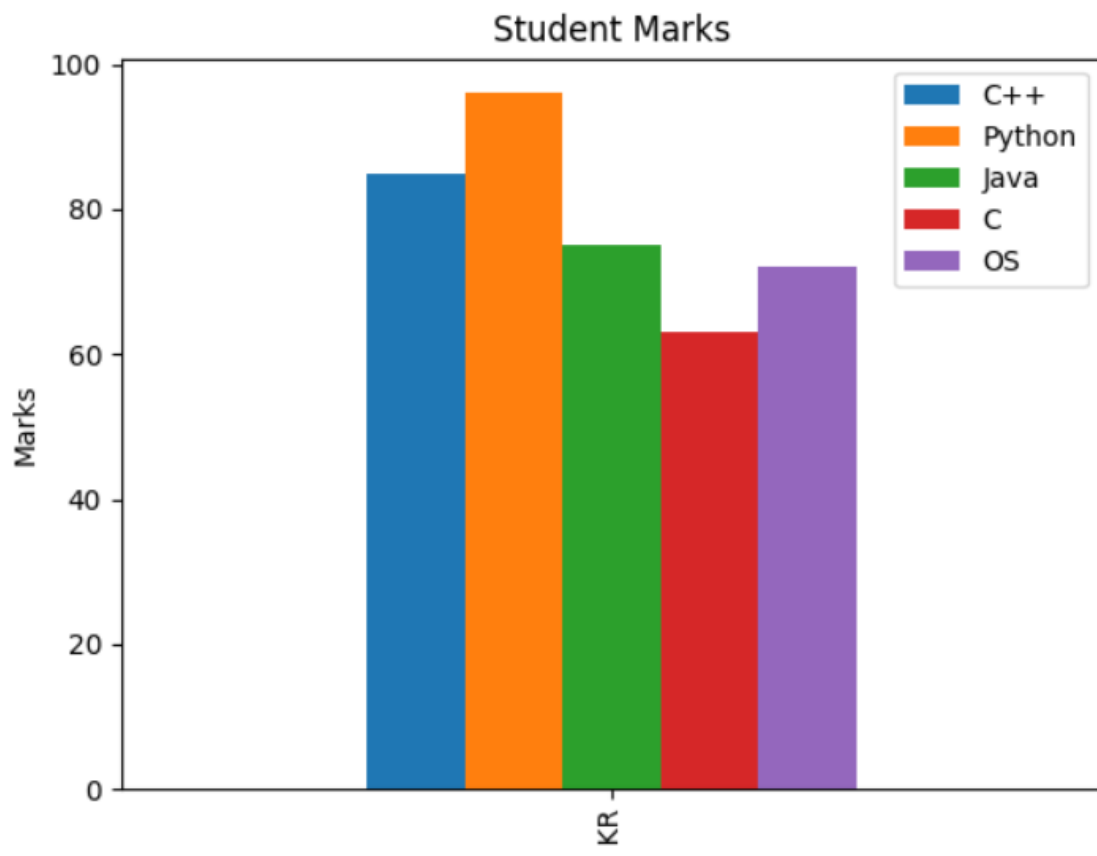
Analyze Result

Success

i

Student data added

OK



```
# Personal Expense Tracker

import tkinter as tk
from tkinter import messagebox
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Store expense data
data = []

# Function to add expense
def add_expense():
    category = category_entry.get()
    amount = float(amount_entry.get())

    data.append([category, amount])

    messagebox.showinfo("Success", "Expense Added")

    category_entry.delete(0, tk.END)
    amount_entry.delete(0, tk.END)

# Function to analyze expenses
def analyze():
    if len(data) == 0:
        messagebox.showwarning("No Data", "Add expenses first")
        return

    df = pd.DataFrame(data, columns=["Category", "Amount"])

    amounts = df["Amount"].values

    total = np.sum(amounts)
    avg = np.mean(amounts)

    result_label.config(
        text=f"Total Expense: {total:.2f}\nAverage Expense: {avg:.2f}"
```

```

result_label.config(
    text=f"Total Expense: {total:.2f}\nAverage Expense: {avg:.2f}"
)

# Pie Chart
category_sum = df.groupby("Category")["Amount"].sum()

plt.figure(figsize=(6,6))
plt.pie(category_sum, labels=category_sum.index, autopct='%1.1f%%')
plt.title("Expense Distribution")
plt.show()

# GUI Window
root = tk.Tk()
root.title("Personal Expense Tracker")
root.geometry("350x300")

# Category input
tk.Label(root, text="Expense Category").pack()
category_entry = tk.Entry(root)
category_entry.pack()

# Amount input
tk.Label(root, text="Amount").pack()
amount_entry = tk.Entry(root)
amount_entry.pack()

# Buttons
tk.Button(root, text="Add Expense", command=add_expense).pack(pady=10)
tk.Button(root, text="Analyze Expenses", command=analyze).pack(pady=10)

# Result display
result_label = tk.Label(root, text="")
result_label.pack(pady=20)

root.mainloop()

```

Expense Category

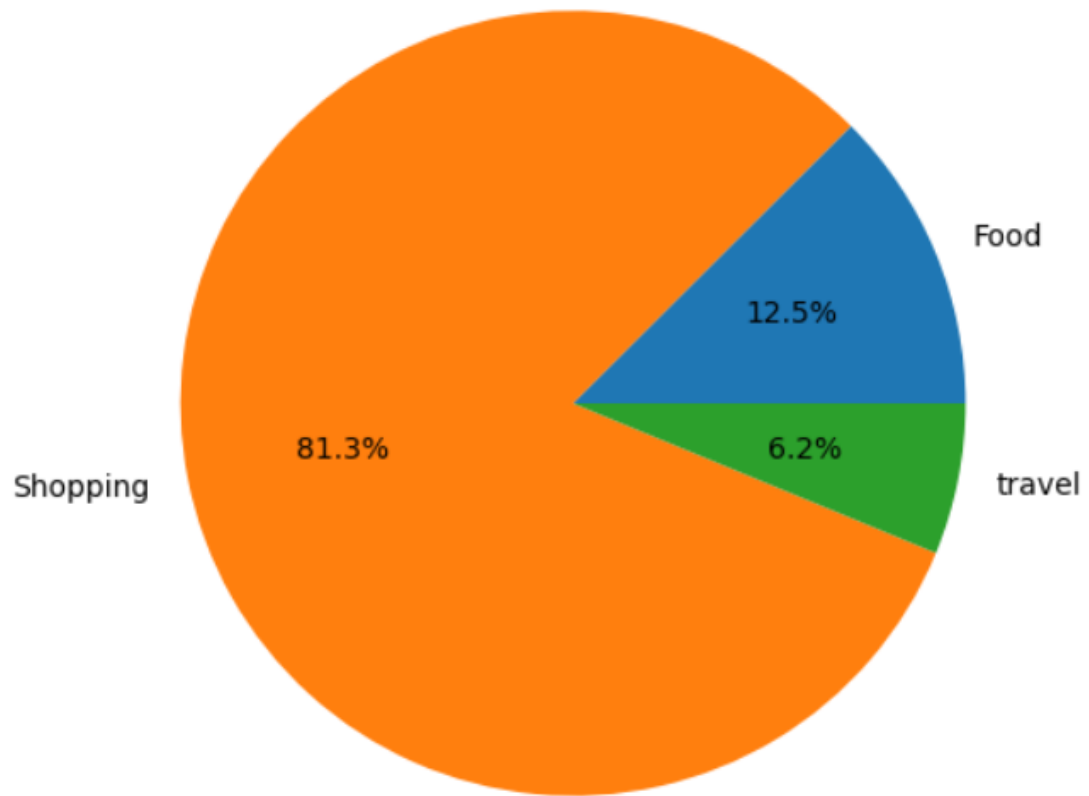
Amount

Add Expense

Analyze Expenses

Total Expense: 3197.00
Average Expense: 1065.67

Expense Distribution



```
# Library Book Data Analyzer

import tkinter as tk
from tkinter import messagebox
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Data storage
data = []

# Add book data
def add_book():
    name = book_entry.get()
    category = category_entry.get()
    copies = int(copies_entry.get())

    data.append([name, category, copies])

    messagebox.showinfo("Success", "Book added successfully")

    book_entry.delete(0, tk.END)
    category_entry.delete(0, tk.END)
    copies_entry.delete(0, tk.END)

# Analyze data
def analyze():
    if len(data) == 0:
        messagebox.showwarning("No Data", "Add books first")
        return

    df = pd.DataFrame(data, columns=["Book", "Category", "Copies"])

    copies = df["Copies"].values

    total = np.sum(copies)
    avg = np.mean(copies)

    result_label.config(
        text=f"Total Books: {total}\nAverage Copies: {avg:.2f}"
    )

# Category-wise bar chart
category_books = df.groupby("Category")["Copies"].sum()
```

```

# Category-wise bar chart
category_books = df.groupby("Category")["Copies"].sum()

category_books.plot(kind="bar")

plt.title("Books by Category")
plt.xlabel("Category")
plt.ylabel("Number of Copies")

plt.xticks(rotation=90)

plt.show()

# GUI window
root = tk.Tk()
root.title("Library Book Data Analyzer")
root.geometry("350x300")

# Inputs
tk.Label(root, text="Book Name").pack()
book_entry = tk.Entry(root)
book_entry.pack()

tk.Label(root, text="Category").pack()
category_entry = tk.Entry(root)
category_entry.pack()

tk.Label(root, text="Number of Copies").pack()
copies_entry = tk.Entry(root)
copies_entry.pack()

# Buttons
tk.Button(root, text="Add Book", command=add_book).pack(pady=10)
tk.Button(root, text="Analyze Data", command=analyze).pack(pady=10)

# Result label
result_label = tk.Label(root, text="")
result_label.pack(pady=20)

root.mainloop()

```

Book Name

Category

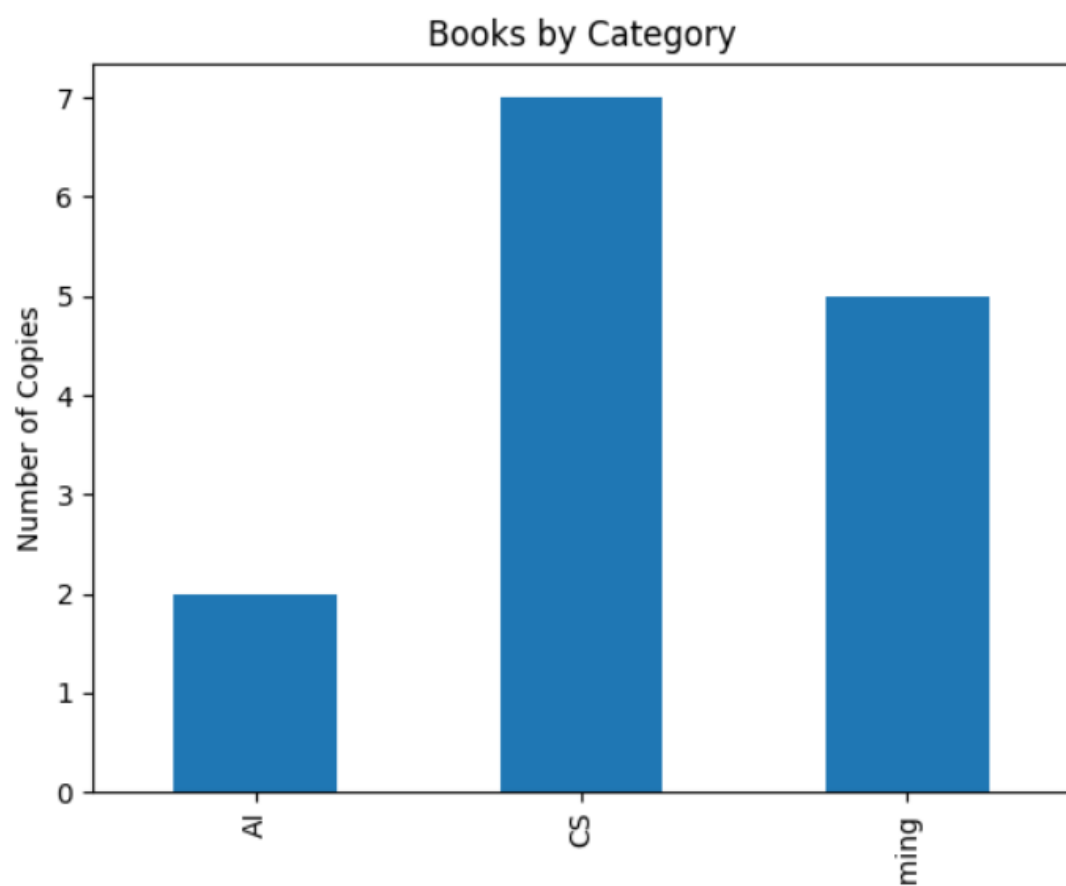
Number of Copies

Add Book

Analyze Data

Total Books: 14

Average Copies: 3.50



```

# Quiz Result Analyzer

import tkinter as tk
from tkinter import messagebox
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Correct answers for quiz
correct_answers = ["A", "B", "C", "D", "A"]

# Data storage
data = []

# Function to add student answers
def submit_answers():
    name = name_entry.get()

    ans1 = q1_entry.get().upper()
    ans2 = q2_entry.get().upper()
    ans3 = q3_entry.get().upper()
    ans4 = q4_entry.get().upper()
    ans5 = q5_entry.get().upper()

    answers = [ans1, ans2, ans3, ans4, ans5]

    # Calculate score
    score = sum([1 for i in range(len(correct_answers)) if answers[i] == correct_answers[i]])

    data.append([name, score])

    messagebox.showinfo("Submitted", f"{name}'s Score: {score}/5")

    # Clear fields
    name_entry.delete(0, tk.END)
    q1_entry.delete(0, tk.END)
    q2_entry.delete(0, tk.END)
    q3_entry.delete(0, tk.END)
    q4_entry.delete(0, tk.END)
    q5_entry.delete(0, tk.END)

# Analyze results
def analyze():
    if len(data) == 0:
        messagebox.showwarning("No Data", "Enter quiz results first")

```

```

    messagebox.showwarning("No Data", "Enter quiz results first")
    return

df = pd.DataFrame(data, columns=["Name", "Score"])

scores = df["Score"].values

avg = np.mean(scores)
highest = np.max(scores)
lowest = np.min(scores)

result_label.config(
    text=f"Average Score: {avg:.2f}\nHighest Score: {highest}\nLowest Score: {lowest}"
)

# Chart
df.set_index("Name").plot(kind="bar")

plt.title("Quiz Scores")
plt.ylabel("Score")
plt.xticks(rotation=90)

plt.show()

# GUI
root = tk.Tk()
root.title("Quiz Result Analyzer")
root.geometry("350x400")

tk.Label(root, text="Student Name").pack()
name_entry = tk.Entry(root)
name_entry.pack()

tk.Label(root, text="Q1 Answer").pack()
q1_entry = tk.Entry(root)
q1_entry.pack()

tk.Label(root, text="Q2 Answer").pack()
q2_entry = tk.Entry(root)
q2_entry.pack()

tk.Label(root, text="Q3 Answer").pack()
q3_entry = tk.Entry(root)
q3_entry.pack()

tk.Label(root, text="Q4 Answer").pack()
q4_entry = tk.Entry(root)
q4_entry.pack()

tk.Label(root, text="Q5 Answer").pack()
q5_entry = tk.Entry(root)
q5_entry.pack()

tk.Button(root, text="Submit Answers", command=submit_answers).pack(pady=10)
tk.Button(root, text="Analyze Results", command=analyze).pack(pady=10)

result_label = tk.Label(root, text="")
result_label.pack(pady=20)

root.mainloop()

```

Student Name

Q1 Answer

Q2 Answer

Q3 Answer

Q4 Answer

Q5 Answer

Submit Answers

Analyze Results

Average Score: 3.33

Highest Score: 5

Lowest Score: 2

